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MILITARY SPECIFICATION

AMPLIFIER-OSCILLATOR GROUP AN/SIA-118

1. SCOPE

1.1 This specification covers the detailed requirements for an audio amplifier-oscillator group AN/SIA-118 for flight deck announcing system, circuit 5MC.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-62 - Capacitors, Fixed, Electrolytic (DC Aluminum, Dry Electrolytic, Polarized) General Specification for.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for.
MIL-C-3965 - Capacitors, Fixed, Electrolytic (Tantalum) General Specification for.
MIL-A-21577 - Amplifiers, Audio Frequency; and Amplifier-Control Groups - Shipboard Announcing.

STANDARDS

MILITARY

MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-701 - Preferred and Guidance List of Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Amplifier-oscillator group AN/SIA-118 shall be in accordance with all applicable paragraphs of MIL-A-21577 and as specified herein. Whenever a requirement of MIL-A-21577 conflicts with a requirement of this specification, the requirement of this specification shall govern.

3.2 Equipment complement shall consist of the following:

(a) One common enclosure, bulkhead mounted, amplifier control rack, maximum dimensions not to exceed 36 inches in height, 24 inches wide, and 21 inches deep containing:

2 Input preamplifier modules.
2 Alarm output preamplifier modules.
2 Power supply modules.
2 Oscillator modules (collision alarm).
2 Oscillator modules (chemical alarm).
2 Oscillator modules (flight crash).
2 Oscillator modules (flight warning).
All necessary switches, meters, relays, wiring, terminals, and other components required to control six amplifier assemblies.

3.3 Modular construction techniques with quick disconnect features shall be employed in the design of these assemblies to permit easy removal of the modules for servicing.

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3.4 All modules shall be mounted in a vertical position to facilitate the unrestricted flow of air between modules for ventilation purposes.

3.5 Capacitors.- Tantalum electrolytic capacitors conforming to MIL-C-3965 shall be used in preamplifiers and oscillator assemblies. Plug-in type electrolytic capacitors may be used in power supply modules. These capacitors shall be in accordance with MIL-C-62.

3.6 Wiring.- Wiring employed shall have insulation type T as shown in the applicable table of MIL-I-983.

3.7 Semiconductor devices.- Semiconductor devices shall be used exclusively and shall be selected from MIL-STD-701.

3.8 Forced air cooling.- Forced air cooling will not be permitted.

3.9 Time constant.- Under the conditions specified in the overload limiter characteristic requirements of MIL-A-21577, the preamplifier output voltage shall return to the value specified in the overload limiter characteristic requirements within 0.025 second after the sudden increase in input signal voltage above reference level.

3.10 Printed wiring.- Where printed wiring is utilized, it shall be in accordance with type C lead termination of MIL-STD-275.

3.11 Test facilities.- A common voltmeter with switching facilities shall be provided on the control rack to determine proper power supply voltage for amplifier-oscillators and preamplifiers. The meter calibration and circuit component values shall be such that the operating condition of the amplifiers or amplifier-oscillators can be determined directly from the meter without reference to tables or charts.

3.12 Interstage input circuit.- Interstage input circuit will not be required.

3.13 Generated signals.- The amplifier-oscillator modules shall be capable of generating the following alarm signals:

Signal number	Signal function
1	Collision alarm
4	Chemical alarm
2	Flight crash alarm
6	Flight warning alarm

3.13.1 Alarm signals numbers 3, 5, 7, and 8 will not be required. Electronic timing shall be provided on circuits where timing is required.

3.14 Output level.- The amplifier-oscillator shall be capable of maintaining an output level of not less than 10 volts into a 600-ohm resistive load for amplifier input. This voltage shall also be used as a test signal source as specified in MIL-A-21577.

3.15 Adjustment of output level.- A screwdriver type adjustment with shaft-locking nut shall be provided in the amplifier-oscillator or in combination with the associated control circuits to permit independent adjustment of the amplifier-oscillator output level for the alarm signal into the amplifier input.

3.16 Input facilities.- Amplifier-Oscillator AN/SIA-118 shall have the following input facilities:

Microphone station number	Microphone station type	Circuits controlled
1 through 6	IC/MSB-2	5 MC

3.17 Local loudspeaker cutout circuit.- A cutout circuit shall be included for each microphone station to disconnect automatically local loudspeakers when the microphone is in use. The cutout circuit shall consist of a normally closed relay which shall operate to open the audio circuit supplying the local loudspeakers when the microphone press-to-talk switch is operated. These relays shall be installed in the amplifier control rack. Six local cutout circuits are required.

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3.18 Output facilities.- Amplifier-Oscillator AN/SIA-118 shall have the following output facilities:

Loudspeaker group number	Loudspeaker group designation	Circuit
1 through 12	Flight deck	5 MC

Circuit 5MC output facilities shall be twelve groups. The circuit 5MC loudspeakers are divided into twelve groups one of which is connected to the output of each 500-watt amplifier assembly.

3.19 Input and output metering and switching shall apply as specified in 3.20.

3.20 Combined metering.- A common volume indicator meter shall be provided for input and output metering. This meter shall indicate zero db at rated input and output power.

3.21 Test input level control.- Test input level control will not be required.

3.22 Microphone press-to-talk switches.- Reduction in output of circuit SE, ship's entertainment system, following operation of a press-to-talk switch shall not apply to the 5MC system.

3.23 Visual alarm.- Operation of alarm signals numbers 1, 4, 2, and 6 shall initiate operation of the visual alarm system.

3.24 Circuit SE muting.- Operation of any alarm contact shall energize a circuit that is capable of muting circuit SE. This circuit shall be capable of making and breaking a resistive load of 7 amperes at 115 volts, 60 cycles, single phase.

3.25 Alarm signal priorities.- The order of priority for alarm signals shall be as follows:

(a) Collision alarm.
(b) Chemical attack alarm.
(c) Flight crash alarm.
(d) Flight warning alarm.

The oscillator modules shall be provided with a device to prevent unauthorized or accidental changes in alarm signal priorities. This device shall allow intentional change in alarm priority by simple mechanical change.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling, examination, and tests shall be as specified in MIL-A-21577.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in MIL-A-21577.

6. NOTES

6.1 The notes specified in MIL-A-21577 are applicable to this specification.

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